Laboratory Record

22CD019- MULTIMEDIA AND ANIMATION LABORATORY (2 0 2 3)

B.E. DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

Semester VI

2025-2026

Submitted by

Reg. No. : ----------

Name : ----------

BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

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DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

BANNARI AMMAN
INSTITUTE OF TECHNOLOGY
Stay Ahead

BONAFIDE CERTIFICATE

This is a Certified Bonafide Record Book of Mr. /Ms. ……..
Register number …… submitted for ……..
….....laboratory during the academic year 2024-2025.

Staff in-charge Head of the Department

Course Objectives:

- Understand the basic knowledge of multimedia Systems and related technologies.
- Learn about multimedia elements in a comprehensive way.
- Understand the basics of digital 2D animation to create story and multimedia production.
- Design the technical and artistic skills to produce 3D animations.

Programme Outcomes (POs):

- PO1: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- PO5: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PSO2: Apply the power of computing and digital media tools to provide solutions to challenging interactive technologies.

Course Outcomes (COs):

1. Analyze the usage of the multimedia elements for computer Image Processing.
2. Analyze the different encode and decode techniques.
3. Apply the author 2D and 3D creative and interactive presentations for different target multimedia applications.
4. Apply 2D animation and develop the storyboards.
5. Create and animate the 3D models using software tools.

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EXPERIMENT – 1

Image Editing and Manipulation: Basic Operations Using Image Editing Software

1. Aim

To perform basic image editing and manipulation operations using image editing software such as Adobe Photoshop or GIMP.

2. Software Required

- Adobe Photoshop / GIMP
- Computer system
- Sample image file (JPEG/PNG format)

3. Theory

Image editing is the process of modifying digital images to improve their appearance or to extract useful information. Image manipulation includes operations like cropping, resizing, adjusting brightness/contrast, rotating, flipping, and applying filters.

Basic image editing operations include:

1. Opening and saving images
2. Cropping
3. Resizing
4. Rotating and flipping
5. Adjusting brightness and contrast
6. Applying filters
7. Adding text

These operations are widely used in multimedia, graphic design, web design, and digital photography.

4. Procedure

Step 1: Open the Image

1. Launch the image editing software.
2. Click File → Open.
3. Select the sample image.

Output:

Original image is displayed in the workspace.

Step 2: Crop the Image

1. Select the Crop Tool.
2. Drag over the required area.
3. Press Enter to apply.

Output:

Unwanted portions are removed; selected area remains.

Step 3: Resize the Image

1. Click Image → Image Size.
2. Change width and height values.
3. Click OK.

Output:

Image size is changed according to specified dimensions.

Step 4: Rotate the Image

1. Click Image → Rotate.
2. Choose 90° clockwise / 180° option.

Output:

Image orientation is changed.

Step 5: Adjust Brightness and Contrast

1. Click Image → Adjustments → Brightness/Contrast.
2. Move sliders as required.
3. Click OK.

Output:

Image brightness and contrast are enhanced.

Step 6: Apply Filter

1. Click Filter menu.
2. Select a filter (e.g., Blur, Sharpen).
3. Apply the filter.

Output:

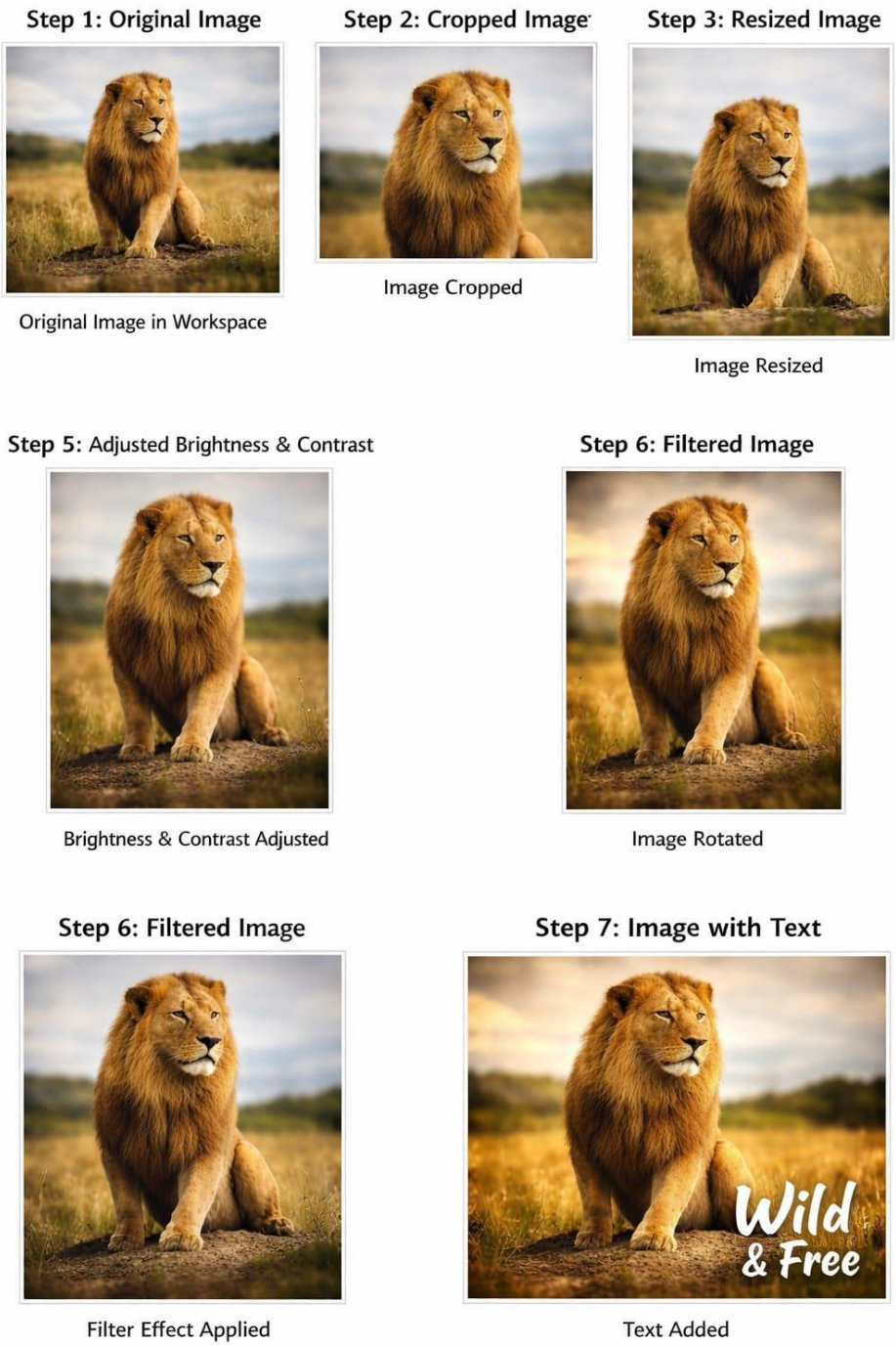
Image appearance changes based on selected filter.

Step 7: Add Text

1. Select Text Tool.
2. Click on the image.
3. Type required text.

Output:

Text appears on the image.



5. Result

Thus, basic image editing and manipulation operations were successfully performed using image editing software.

EXPERIMENT – 2

Implementation of Audio and Video Editing Techniques

1. Aim

To implement basic audio and video editing techniques using multimedia editing software.

2. Software Required

- Audacity (for Audio Editing)
- Adobe Premiere Pro / Filmora (for Video Editing)
- Computer System
- Sample audio (.mp3/.wav) and video (.mp4/.avi) files

3. Theory

Audio and video editing are important components of multimedia systems.

Audio Editing

Audio editing includes:

- Trimming unwanted portions
- Noise removal
- Volume adjustment
- Adding effects (Echo, Fade In/Out)
- Mixing multiple tracks

Video Editing

Video editing includes:

- Cutting and trimming clips
- Adding transitions
- Inserting background music
- Adding text/titles
- Applying filters and color correction

These techniques are widely used in filmmaking, YouTube content creation, advertising, and digital media production.

4. Procedure

PART A – Audio Editing (Using Audacity)

Step 1: Import Audio

- Open Audacity
- Click **File → Import → Audio**
- Select sample audio file

Output: Audio waveform appears on timeline.

Step 2: Trim Audio

- Select unwanted portion
- Click **Delete**

Output: Unwanted section removed.

Step 3: Reduce Noise

- Select noise sample
- Click **Effect → Noise Reduction**
- Apply effect

Output: Background noise reduced.

Step 4: Apply Fade In / Fade Out

- Select beginning portion → Apply Fade In
- Select ending portion → Apply Fade Out

Output: Smooth audio start and ending.

Step 5: Export Edited Audio

- Click **File → Export → MP3**

Output: Edited audio file saved successfully.

PART B – Video Editing (Using Premiere Pro / Filmora)

Step 1: Import Video

- Open Video Editing Software
- Click **Import Media**
- Drag video to timeline

Output: Video clip displayed in preview panel.

Step 2: Trim Video

- Use Razor tool
- Cut unwanted portion

Output: Extra scenes removed.

Step 3: Add Transition

- Drag transition (Fade, Dissolve) between clips

Output: Smooth transition between scenes.

Step 4: Add Text / Title

- Click **Text Tool**
- Type title on video

Output: Title appears on video screen.

Step 5: Add Background Music

- Import audio
- Place below video track

Output: Background music added.

Step 6: Export Video

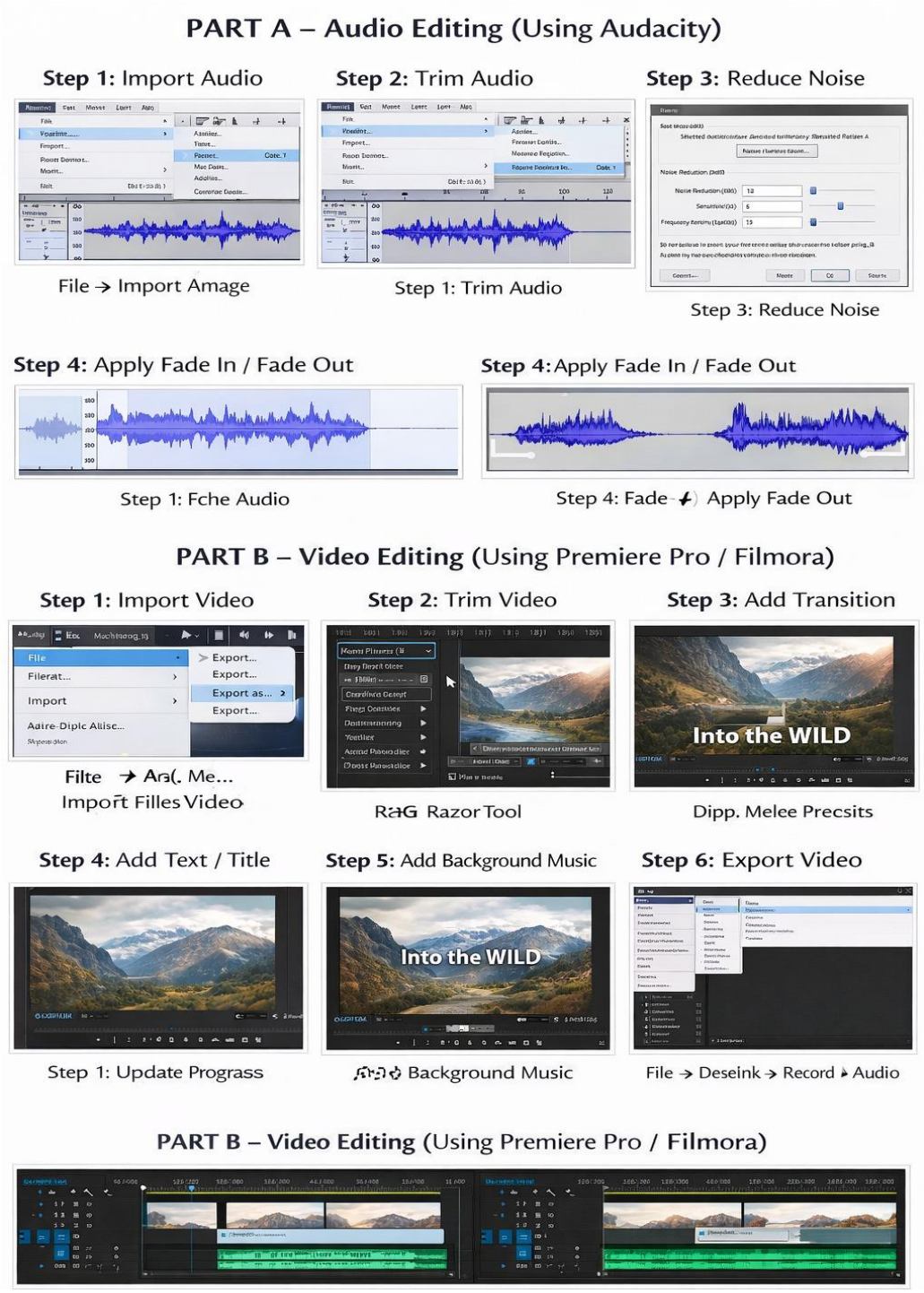
- Click **Export → MP4 format**

Output: Final edited video saved successfully.

5. Sample Output Image

Below is a sample representation of edited media outputs (Audio waveform + Video editing timeline):

(Represents trimming, enhancement, and editing techniques applied.)



6. Result

Thus, the implementation of basic audio and video editing techniques was successfully performed using multimedia editing software.

EXPERIMENT – 3

Sketching of Cartoon Characters

1. Aim

To create and sketch cartoon characters using digital drawing tools in image editing software.

2. Software Required

- Adobe Photoshop / Adobe Illustrator
- Drawing tablet or mouse
- Computer system

3. Theory

Cartoon character sketching is a creative process used in animation, games, and multimedia design. Digital sketching allows easy editing, coloring, resizing, and shading. Cartoon characters are usually drawn with simplified shapes, exaggerated facial expressions, and bold outlines to make them visually appealing.

Basic steps involved in cartoon sketching:

- Rough sketch using basic shapes
- Clean outline
- Coloring
- Adding shadows and highlights

4. Procedure

Step 1: Create New Canvas

1. Open the drawing software.
2. Click **File → New**.
3. Set canvas size (e.g., 800 × 800 pixels).
4. Click **Create**.

Output: Blank canvas created.

Step 2: Rough Sketch

1. Select **Brush Tool**.
2. Draw basic shapes (circle for head, oval for body).
3. Sketch facial features lightly.

Output: Rough outline of cartoon character drawn.

Step 3: Clean Outline

1. Reduce opacity of rough sketch layer.
2. Create a new layer.
3. Draw clean outlines using Pen/Brush tool.

Output: Neat and clear cartoon outline.

Step 4: Coloring

1. Select **Paint Bucket / Brush Tool**.
2. Fill colors for skin, dress, hair, etc.

Output: Cartoon character colored.

Step 5: Add Details

1. Add eyes, mouth, shadows, and highlights.
2. Adjust thickness of lines.

Output: Final cartoon character sketch completed.

Step 6: Save the Sketch

1. Click **File → Save As**.
2. Save in PNG/JPEG format.

Output: Cartoon character image saved successfully.

5. Sample Output Description

- A cartoon character with:
  - Rounded head
  - Large expressive eyes
  - Simple body structure
  - Bright colors
- Clean outlines and smooth shading.

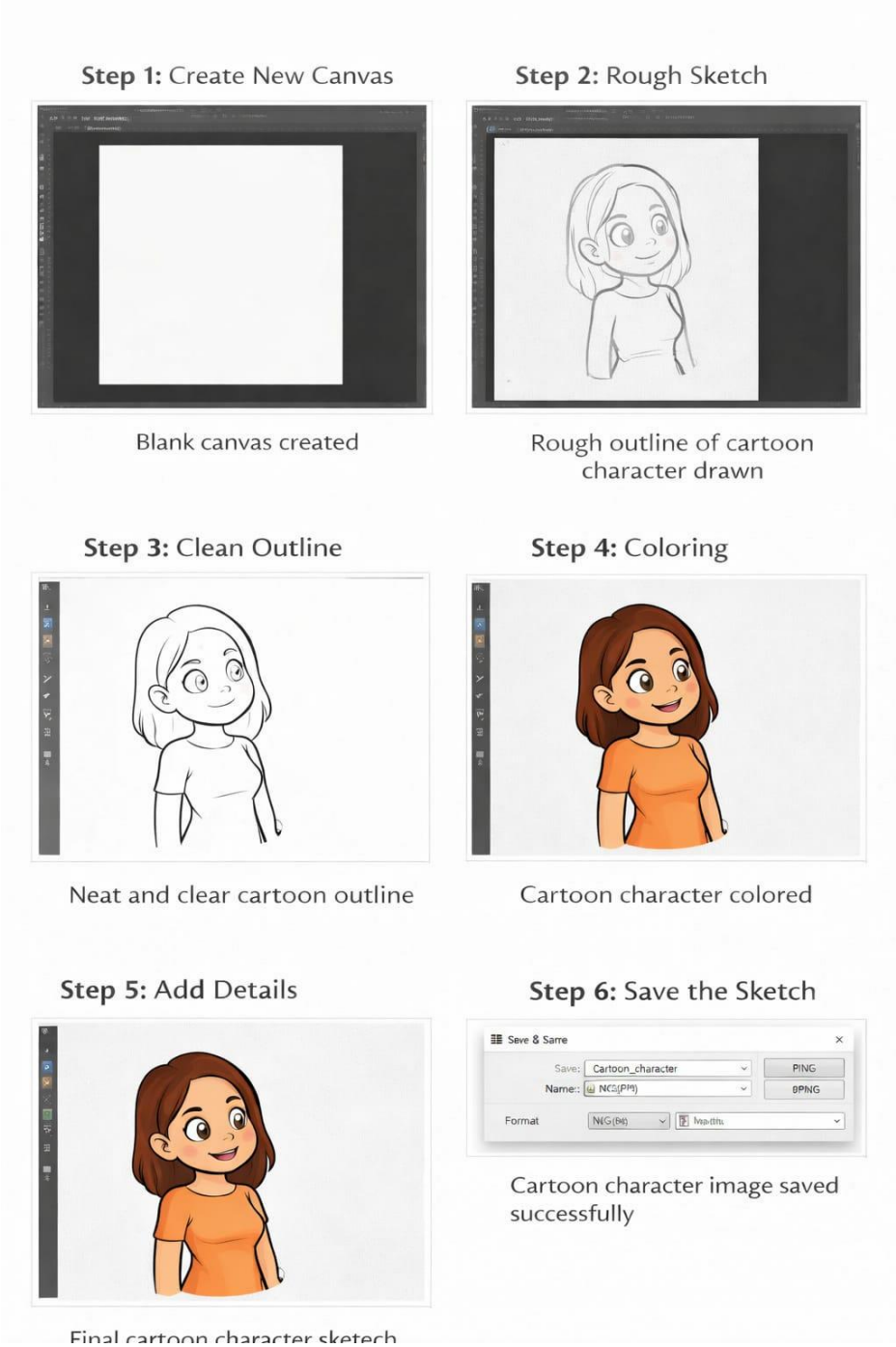
6. Sample Output Image

(Example representation of cartoon sketch)

Cartoon Character Sketch

- Clear outline
- Bright colors
- Expressive face

(You may paste your drawn cartoon image here in the lab record.)



7. Result

Thus, a cartoon character was successfully sketched using digital drawing tools.

EXPERIMENT – 4

Design 2D Logo Using Image Editing Tool

1. Aim

To design a 2D logo using an image editing tool.

2. Software Required

- Adobe Photoshop / Adobe Illustrator / GIMP
- Computer System

3. Theory

A logo is a graphical symbol that represents a company, brand, institution, or product. A 2D logo is designed using shapes, colors, text, and symbols without three-dimensional effects.

Key principles of logo design:

- Simplicity
- Scalability
- Relevance
- Balance
- Color harmony

Image editing tools provide features such as:

- Shape tools (Circle, Rectangle, Polygon)
- Text tool
- Gradient tool
- Layer management
- Color adjustments

4. Procedure

Step 1: Create New Canvas

1. Open the image editing software.
2. Click **File → New**.
3. Set canvas size (e.g., 800 × 800 pixels).
4. Choose white or transparent background.

Output: Blank canvas created.

Step 2: Create Basic Shape

1. Select **Shape Tool**.
2. Draw a circle or rectangle.
3. Choose fill color.

Output: Base shape of logo created.

Step 3: Add Text

1. Select **Text Tool**.
2. Type brand or company name.
3. Adjust font size, color, and alignment.

Output: Text added to logo.

Step 4: Add Design Elements

1. Insert icons or additional shapes.
2. Apply gradient or solid colors.
3. Arrange layers properly.

Output: Logo design enhanced with graphical elements.

Step 5: Final Adjustments

1. Align objects properly.
2. Adjust brightness/contrast if required.
3. Check spacing and balance.

Output: Final 2D logo design ready.

Step 6: Save the Logo

1. Click **File → Save As**.
2. Save as PNG/JPEG format.

Output: 2D logo saved successfully.

5. Sample Output Description

Example Logo Design:

- Circular base with blue background
- White initials in center
- Simple minimal design
- Balanced alignment

- Clean and professional appearance

6. Sample Output Representation

Example:

🟢 Blue circular logo

White text: *"KS Design"*

Simple flat 2D style

(You can paste your designed logo image here in your lab record.)

7. Result

Thus, a 2D logo was successfully designed using an image editing tool.

EXPERIMENT – 5

Creating GIF Animated Images in 2D Animation

1. Aim

To create a GIF animated image using 2D animation techniques.

2. Software Required

- Adobe Photoshop

  or

- GIMP
- Computer System

3. Theory

GIF (Graphics Interchange Format) is an image format that supports animation by displaying multiple frames in sequence.

2D animation is the process of creating movement in a two-dimensional space by showing a sequence of images (frames).

Key Concepts:

- **Frame** – Each individual image in animation
- **Timeline** – Displays frames in sequence
- **Frame Delay** – Time between frames
- **Looping** – Repeating animation continuously

GIF animations are widely used in:

- Social media
- Websites
- Advertisements
- Multimedia projects

4. Procedure

Method Using Photoshop

Step 1: Create New File

1. Open Photoshop.
2. Click **File → New**.

3. Set canvas size (e.g., 500 × 500 pixels).

Output: Blank canvas created.

Step 2: Create Frames

1. Draw an object (example: bouncing ball).
2. Duplicate layer.
3. Move object slightly in second layer.
4. Repeat for multiple positions.

Output: Multiple layers representing movement.

Step 3: Open Timeline Panel

1. Click **Window → Timeline**.
2. Select **Create Frame Animation**.

Output: Frame timeline displayed.

Step 4: Create Animation Frames

1. Convert layers into frames.
2. Set frame delay (e.g., 0.2 seconds).
3. Choose **Loop: Forever**.

Output: Animation preview shows movement.

Step 5: Export as GIF

1. Click **File → Export → Save for Web**.
2. Select GIF format.
3. Click Save.

Output: Animated GIF file created successfully.

5. Sample Animation Example

Example: **Bouncing Ball Animation**

Frame 1 – Ball at top

Frame 2 – Ball moving down

Frame 3 – Ball at bottom

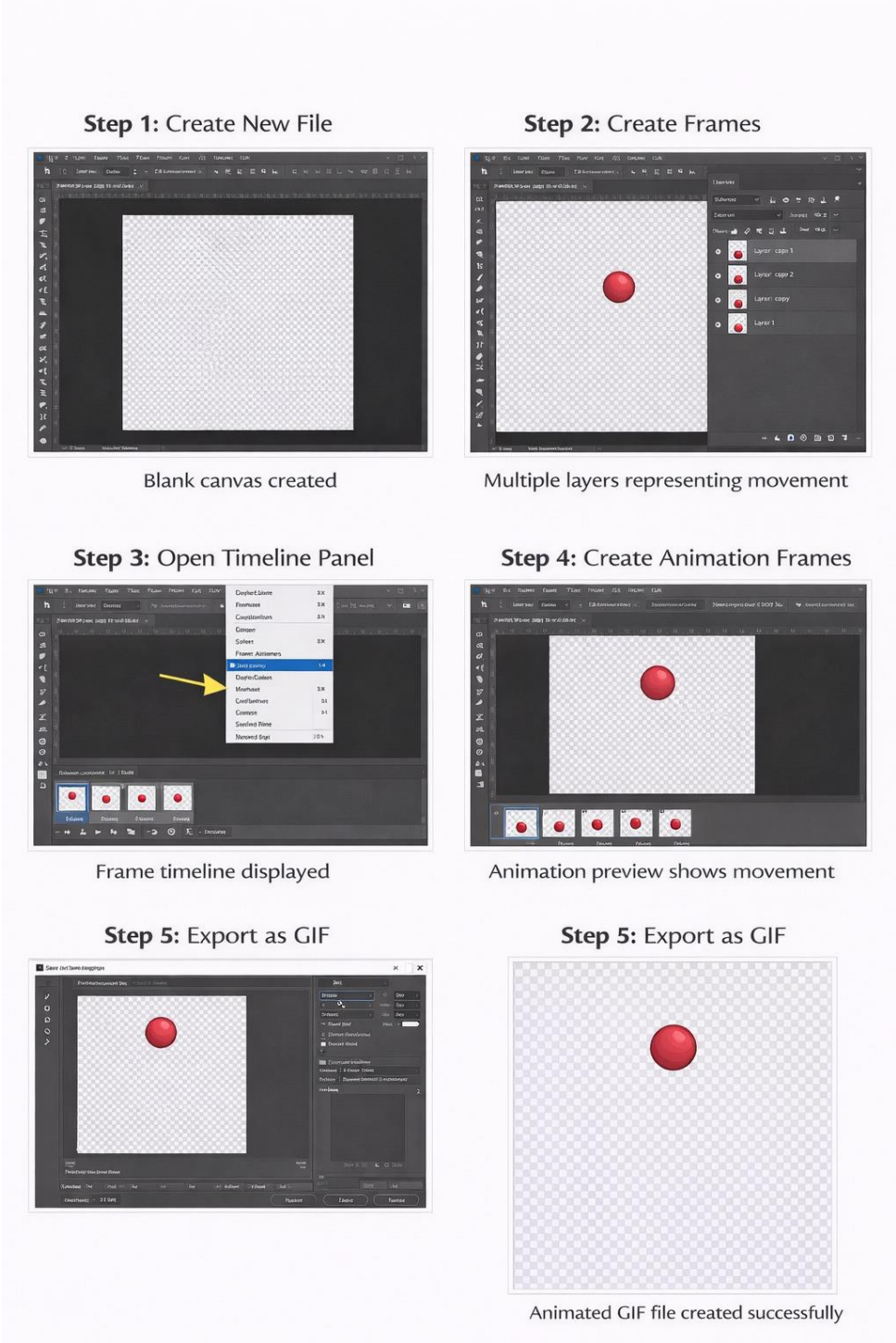
Frame 4 – Ball moving up

When played continuously → Ball appears to bounce.

6. Sample Output Description

- 5 frames used

- Frame delay: 0.2 seconds
- Loop: Forever
- Smooth 2D motion visible
- GIF file size optimized



7. Result

Thus, a GIF animated image was successfully created using 2D animation techniques.

EXPERIMENT – 6

Exploring the Interface of 3D application

Aim:

To study and understand the user interface, tools, and basic operations of a 3D application.

Software Required

- Blender (latest version recommended)
- Minimum 4GB RAM system

Theory

A 3D application is used to create models, animations, and visual effects. The interface consists of different panels and tools that help users create and manipulate 3D objects.

Key components include:

- Viewport
- Toolbars
- Panels
- Timeline
- Properties

Procedure (Step-by-Step)

1. Open the Software

- Launch **Blender**
- You will see the default scene with:
  - Cube
  - Camera
  - Light

2. Understand the Main Interface

1. 3D Viewport

- Main working area
- Used to create and edit objects
- Controls:
  - Rotate: Middle Mouse Button
  - Zoom: Scroll wheel
  - Pan: Shift + Middle Mouse

2. Toolbar (Left Side)

- Shortcut key: Press **T**
- Tools available:
  - Move

- Rotate
- Scale
- Transform

3. Outliner (Top Right)

- Shows all objects in the scene
- Example:
  - Cube
  - Camera
  - Light

4. Properties Panel (Right Side)

- Shortcut key: Press **N**
- Contains:
  - Object properties
  - Material settings
  - Render settings

5. Timeline (Bottom)

- Used for animation
- Frame control (Play, Pause)

6. Menu Bar (Top)

Options include:

- File (Save, Open)
- Edit
- Render
- Window
- Help

Basic Operations

➢ **Add Object**

- Press **Shift + A**
- Select Mesh → Cube / Sphere

➢ **Move Object**

- Press **G**
- Move using mouse

➢ **Rotate Object**

- Press **R**

➢ **Scale Object**

- Press **S**

➢ **Delete Object**

- Select object → Press **X**

✓ **Changing View Modes**

- Wireframe → Z → Wireframe
- Solid → Z → Solid
- Rendered → Z → Rendered

Saving Your Work

- Go to File → Save As
- Choose location and save as .blend

Result

The interface of the 3D application was successfully explored. Different panels, tools, and basic object operations were understood.

EXPERIMENT 7:

Title:

Creating Different Types of Materials and Shading

1. Aim

To create and apply different types of materials and shading effects using a 3D application such as Blender.

2. Software Required

- Blender
- Computer system

3. Theory

Materials and shading are essential in 3D graphics to define how an object looks when light interacts with it.

- **Material**: Defines the surface properties like color, texture, reflectivity, and transparency.
- **Shading**: Determines how light and shadow appear on the object.

Types of shading:

1. Flat Shading
2. Smooth Shading
3. Wireframe Shading
4. Rendered Shading

Types of materials include:

- Diffuse (matte surface)
- Glossy (shiny surface)
- Transparent (glass-like)
- Emission (light-emitting)

These are widely used in animation, gaming, and product visualization.

4. Procedure

Step 1: Open the Software

1. Launch **Blender**.
2. Open the default scene.

Output:
Default cube appears in the viewport.

Step 2: Add Objects

1. Press **Shift + A**.
2. Select **Mesh → Sphere / Cube**.

Output:
New object is added to the scene.

Step 3: Apply Material

1. Select the object.
2. Go to **Material Properties** panel.
3. Click **New**.
4. Change base color.

Output:
Material is applied with selected color.

Step 4: Create Different Materials

a) Diffuse Material

- Keep roughness high
- Low reflectivity

Output:
Object appears dull/matte.

b) Glossy Material

- Increase specular
- Reduce roughness

Output:
Object appears shiny.

c) Transparent Material

- Reduce alpha value
- Use transmission option

Output:
Object appears glass-like.

d) Emission Material

- Increase emission strength
- Choose bright color

Output:
Object emits light.

Step 5: Apply Shading

Flat Shading

1. Right-click object
2. Select **Shade Flat**

Output:
Surface appears faceted.

Smooth Shading

1. Right-click object
2. Select **Shade Smooth**

Output:
Surface appears smooth.

Step 6: Change View Mode

1. Press **Z**
2. Select **Rendered / Material Preview**

Output:
Realistic lighting and shading effects are visible.

EXPERIMENT – 7
Creating Different Types of Materials and Shading

1. Aim
To create and apply different types of materials and shading effects using a 3D application such as Blender.

2. Software Required
- Blender
- Computer system

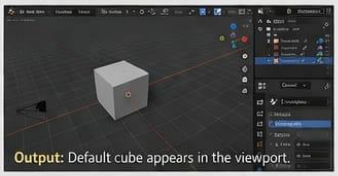
3. Theory
Materials and shading are essential in 3D graphics to define how an object looks when light interacts with it.
- **Material:** Defines the surface properties like color, texture, reflectivity, and transparency.
- **Shading:** Determines how light and shadow appear on the object.

Types of shading:
- Flat Shading
- Smooth Shading
- Wireframe Shading
- Rendered Shading

Types of materials include:
- Diffuse (matte surface)
- Glossy (shiny surface)
- Transparent (glass-like)
- Emission (light-emitting)

These are widely used in animation, gaming, and product visualization.

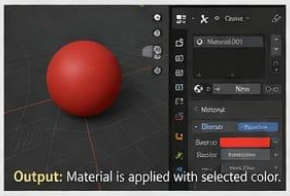
4. Procedure (Step-by-Step)
Step 1: Open the Software
- Launch Blender.
- Open the default scene.


Output: Default cube appears in the viewport.

Step 2: Add Objects
- Press **Shift + A**.
- Select **Mesh → Sphere / Cube.**

Output: New object is added to the scene.

Step 3: Apply Material
- Select the object.
- Go to Material Properties panel.
- Click **New.**
- Change base color.


Output: Material is applied with selected color.

Step 4: Create Different Materials
a) Diffuse Material
- Keep roughness high
- Low reflectivity

Output: Object appears dull/matte.

b) Glossy Material
- Increase specular
- Reduce roughness

Output: Object appears shiny.

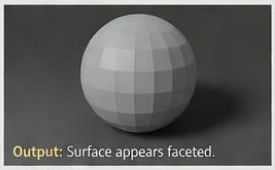
c) Transparent Material
- Reduce alpha value
- Use transmission option

Output: Object appears glass-like.

d) Emission Material
- Increase emission strength
- Choose bright color

Output: Object emits light.

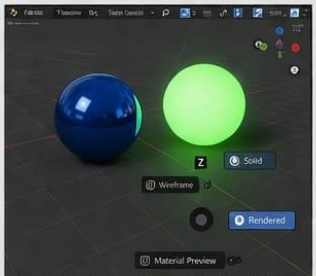
Step 5: Apply Shading
Flat Shading
- Right-click object
- Select Shade **Flat**


Output: Surface appears faceted.

Smooth Shading
- Right-click object
- Select Shade Smooth

Output: Surface appears smooth.

Step 6: Change View Mode
- Press Z
- Select **Rendered / Material Preview**



Output: Realistic lighting and shading effects are visible.

4. Different Materials Created

Diffuse (Matte)	Glossy (Shiny)
Dull, non-reflective surface	Smooth, reflective surface
Transparent (Glass)	**Emission**
Glass-like, see-through	Light-emitting surface

5. Result
Thus, different types of materials and shading techniques were successfully created and applied using Blender.

5. Result

Thus, different types of materials and shading techniques were successfully created and applied using Blender.

EXPERIMENT 8

Create a Simple Walk Cycle Using Character Rigs

Title:

Create a Simple Walk Cycle Using Character Rigs

1. Aim

To create a simple walk cycle animation using character rigs in a 3D application such as Blender.

2. Software Required

- Blender
- Computer system

3. Theory

A walk cycle is a sequence of repeated movements that simulate walking. It is one of the fundamental concepts in character animation.

- **Rigging**: The process of adding a skeleton (bones) to a character model to control its movement.
- **Armature**: The bone structure used for animation.
- **Keyframes**: Frames where specific positions are defined.

Basic steps in a walk cycle include:

1. Contact Position
2. Down Position
3. Passing Position
4. Up Position

These steps create a smooth and continuous walking motion when looped.

4. Procedure

Step 1: Open the Software

1. Launch **Blender**.
2. Open a new project.

Output:
Default scene appears.

Step 2: Add a Character Rig

1. Press **Shift + A**
2. Select**Armature→Human(orRiggedModel)**
   (Or import a pre-rigged character)

Output:
Character with skeleton (rig) appears.

Step 3: Switch to Pose Mode

1. Select the armature
2. Change mode to **Pose Mode**

Output:
Bones become movable for animation.

Step 4: Set Timeline

1. Set timeline frames (e.g., 1 to 40)
2. Enable **Auto Keyframe**

Output:
Animation timeline is ready.

Step 5: Create Key Poses

Frame 1 – Contact Position

- Move legs opposite (one forward, one backward)
- Arms swing opposite to legs

Output:
Initial walking pose is created.

Frame 10 – Down Position

- Lower the body slightly
- Bend front leg

Output:
Body weight shifts downward.

Frame 20 – Passing Position

- One leg passes the other
- Body returns to center

Output:
Mid-step pose is created.

Frame 30 – Up Position

- Raise the body slightly
- Back leg pushes forward

Output:
Body lifts upward.

Frame 40 – Repeat Contact Position

- Copy frame 1 pose

Output:
Cycle completes smoothly.

Step 6: Play Animation

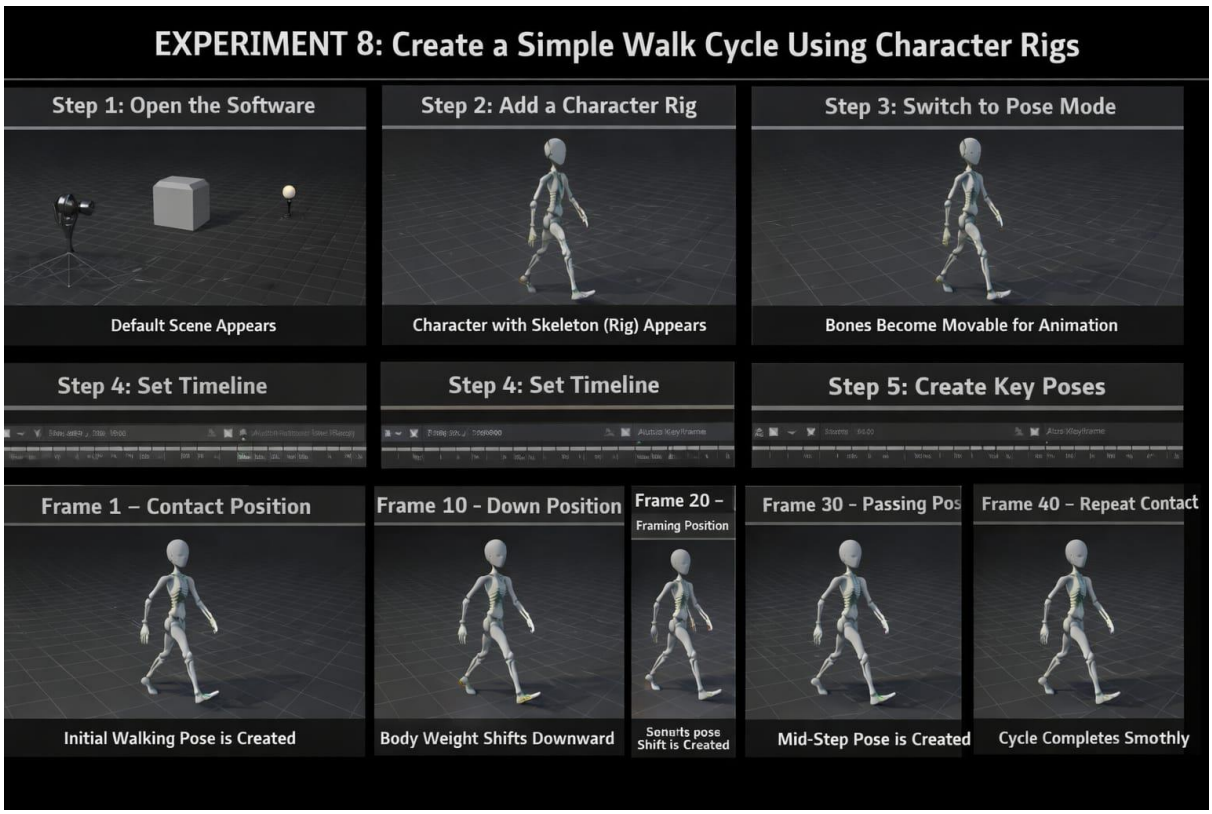
1. Press **Spacebar**
2. Observe the motion

Output:
Character performs a walking animation.

Step 7: Loop the Animation

1. Ensure first and last frames match
2. Set animation to loop

Output:
Continuous walk cycle is achieved.



5. Result

Thus, a simple walk cycle was successfully created using character rigs in Blender.

EXPERIMENT 9

Create a 3-Point Light Setup

Title:

Create a 3-Point Light Setup

1. Aim

To create a 3-point lighting setup in a 3D application such as Blender.

2. Software Required

- Blender
- Computer system

3. Theory

3-point lighting is a standard method used in photography, cinematography, and 3D rendering to illuminate a subject effectively.

It consists of three types of lights:

1. **Key Light**
    - Main and strongest light source
    - Placed at an angle to the subject
2. **Fill Light**
    - Softer light
    - Reduces shadows created by the key light
3. **Back Light (Rim Light)**
    - Placed behind the subject
    - Separates the object from the background

This setup creates depth, realism, and balanced illumination in the scene.

4. Procedure

Step 1: Open the Software

1. Launch **Blender**
2. Open a new project

Output:

Default scene appears with cube, camera, and light.

Step 2: Prepare the Scene

1. Delete the default light (optional)
2. Keep the cube or add a sphere (**Shift + A → Mesh**)

Output:

Object is ready for lighting.

Step 3: Add Key Light

1. Press **Shift + A → Light → Area/Point Light**
2. Position it at a 45° angle to the object
3. Increase power

Output:

Main illumination is created on the object.

Step 4: Add Fill Light

1. Add another light (**Shift + A → Light**)
2. Place it opposite to the key light
3. Reduce intensity

Output:

Shadows are softened.

Step 5: Add Back Light

1. Add a third light
2. Place it behind the object
3. Increase brightness slightly

Output:

Edges of the object are highlighted (rim effect).

Step 6: Adjust Light Settings

1. Modify intensity, color, and distance
2. Adjust shadow softness

Output:

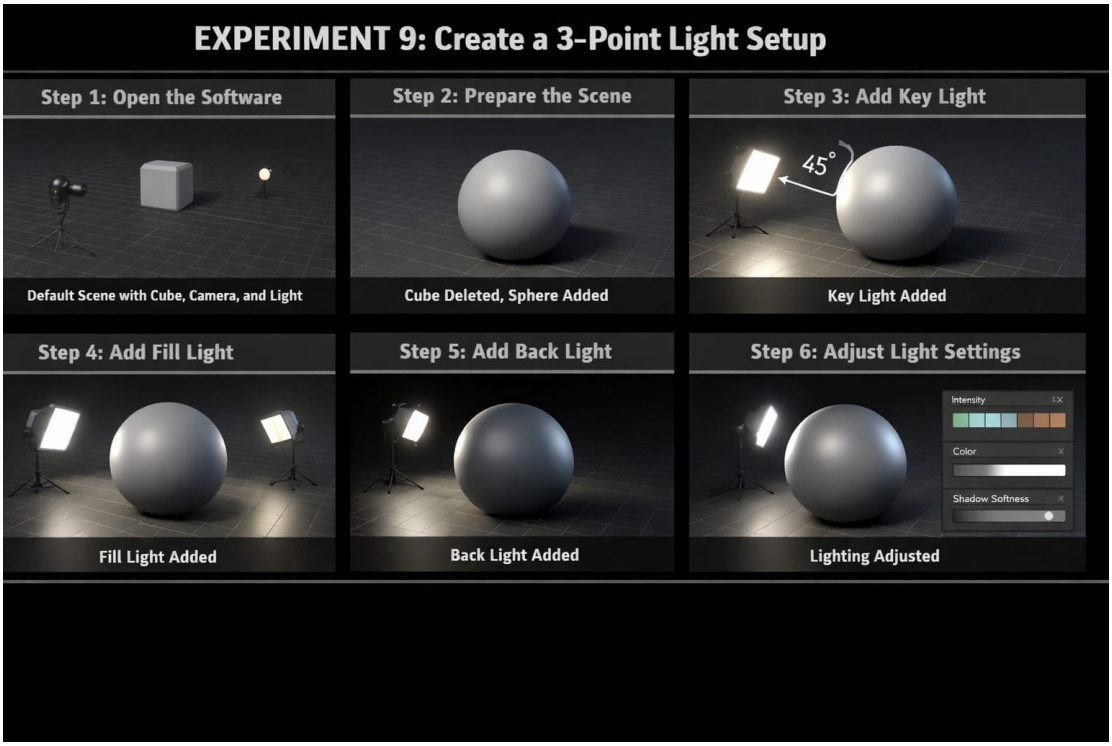
Balanced lighting effect is achieved.

Step 7: Render the Scene

1. Press **Z → Rendered View**
2. Or click **Render → Render Image**

Output:

Final image shows proper 3-point lighting.



5. Result

Thus, a 3-point lighting setup was successfully created using Blender.

EXPERIMENT 10

Create a Particle Simulation

Title:

Create a Particle Simulation

1. Aim

To create a basic particle simulation using a 3D application such as Blender.

2. Software Required

- Blender
- Computer system

3. Theory

A particle system is used in 3D graphics to simulate a large number of small objects such as rain, smoke, fire, dust, or sparks.

- **Particles** are tiny elements emitted from an object.
- **Emitter** is the object that generates particles.
- **Physics** controls particle motion (gravity, velocity, etc.).

Types of particle simulations:

1. Emitter (rain, fire, sparks)
2. Hair (grass, fur)

Particle systems are widely used in animation, visual effects, and games.

4. Procedure

Step 1: Open the Software

1. Launch **Blender**
2. Open a new project

Output:

Default scene appears.

Step 2: Add Emitter Object

1. Select the cube (or add a plane using **Shift + A → Mesh → Plane**)
2. Scale the object if required

Output:

Object is ready to emit particles.

Step 3: Add Particle System

1. Select the object
2. Go to **Particle Properties**
3. Click **+ (Add Particle System)**

Output:

Particles are generated from the object.

Step 4: Adjust Particle Settings

1. Set **Number of particles**
2. Adjust **Lifetime**
3. Modify **Velocity and Direction**

Output:

Particle flow changes according to settings.

Step 5: Apply Physics

1. Enable **Gravity**
2. Adjust force fields if needed

Output:

Particles move realistically.

Step 6: Change Particle Type

- Choose **Emitter** for effects like rain/fire
- Choose **Hair** for grass/fur

Output:

Different particle effects are created.

Step 7: Play Simulation

1. Press **Spacebar**
2. Observe particle motion

Output:

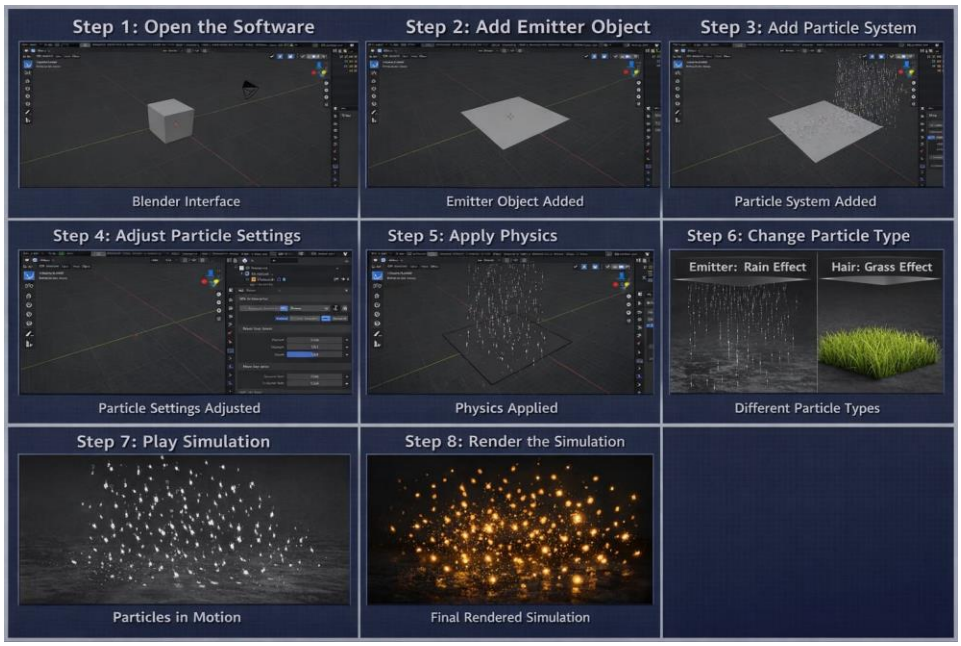
Particles animate over time.

Step 8: Render the Simulation

1. Switch to **Rendered View (Z → Rendered)**
2. Or click **Render → Render Image**

Output:

Final particle simulation is displayed.



5. Result

Thus, a particle simulation was successfully created using Blender.